

A Housing and Additional Children: Evidence from Sibling Composition in the ACS

This appendix asks how the housing a young mother occupies differs when she has an additional child. We use two features of her children that the literature has used as sources of variation in family size. The first is whether her two oldest resident children are recorded at the same age, an age-only proxy for a twin birth. The second is whether her two oldest resident children have the same sex, which Angrist and Evans (1998) show raises the probability of a third birth. Each delivers a causal effect of the additional child only under assumptions we state in Section A.4. Housing is a different outcome from the labor-market outcomes for which these designs were developed, and for the sibling-sex design an awkward one: children of the same sex can share a bedroom, so the composition of a family might change the housing it needs without changing its size.

The two designs study different families at different margins, and they give answers of opposite sign. Mothers whose two oldest children are recorded at the same age live in about a third of a room more than comparable mothers, and are about three percentage points more likely to own their home. Mothers whose two oldest children have the same sex live in slightly *fewer* rooms and fewer bedrooms. Scaled by the first stage, the second design implies that a third child reduces a family’s dwelling by more than a full room. We report this estimate as it is, set out what would have to be true for it to be an effect of the third child, and describe the checks we ran on its measurement.

A.1 Data and the mother sample

The data are the one-year samples of the American Community Survey (ACS) for 2005–2019, covering the fifty states and the District of Columbia, as distributed by IPUMS USA. Each survey year is a separate cross-section of households. A person is identified by the survey year, the sample, the household serial number, and the person’s number within the household; a household is identified by the first three of these. No person is followed across years, so every quantity below is measured once, at the interview.

The ACS does not ask women how many children they have borne; it asks only whether a woman gave birth in the past twelve months, and it records no birth histories. We therefore reconstruct each woman’s resident children from the household roster. IPUMS assigns to each person the person number of a probable mother living in the same household, using relationship, age, and marital information (IPUMS USA, 2026). These links identify social as well as biological relationships: a stepmother or adoptive mother is linked in the same way as a birth mother. We keep every link in which the child’s age is nonnegative and below the mother’s age. The household relationship variable, which records each person’s relationship to the householder rather than to the mother, cannot separate biological from other links and is not used to filter them. Among all valid links in the 2005–2019 records, before any restriction on ages, 91 percent of linked children are recorded as the householder’s child and 4 percent as the householder’s grandchild. Only 1,692 links fail because the recorded child is not younger than the recorded mother.

For each woman we count her linked resident children, m_i , and order them by age, $a_{i1} \geq a_{i2} \geq \dots$, where ages are whole years at the interview. The count m_i is the number of her children living with her, not the number she has borne. Children who have left home, died, or live with another parent are missing, and a woman can be linked to a child she did not bear. Following Angrist and Evans (1998, pp. 452–453), we limit these errors by studying young mothers: the base sample consists of women aged 21 to 35 at the interview whose oldest linked child, if any, is younger than 18. Angrist and Evans additionally drop mothers whose count of resident children differs from their

reported children ever born and mothers whose children’s ages or sexes were imputed by the Census Bureau. The ACS does not collect children ever born, and our extract does not carry imputation flags, so neither restriction is applied here; imputed ages and sexes, if any, enter the instruments as recorded.

Table 1 reports the samples. The 59,046,776 person records in the extract fall to 46,373,936 once the sample is limited to the 2005–2019 one-year files. The base sample contains 4,024,238 women living in 3,783,862 households. It includes women with no linked child, who enter neither design. Because several eligible women can live in one household, household counts are reported alongside counts of mothers throughout.

Table 1: Sample construction

	Count
<i>Source records and base sample</i>	
ACS person records, fifty states and the District of Columbia	59,046,776
of which in the 2005–2019 one-year samples	46,373,936
Base sample: women aged 21–35, positive person weight, no linked child aged 18 or older	4,024,238
Households containing at least one base-sample woman	3,783,862
<i>Same-age design</i>	
Estimation sample: base-sample women whose oldest linked child is aged 0–5	857,607
within it, two or more linked children	325,171
within it, two oldest linked children the same age	16,496
within it, three oldest linked children the same age	388
<i>Same-sex design</i>	
Base-sample women with two or more linked children	1,132,295
among them, two oldest linked children the same age (second child of any age)	32,186
Estimation sample (criteria in Section A.3)	656,986
within it, two boys	173,813
within it, two girls	156,161
within it, boy then girl	164,253
within it, girl then boy	162,759
within it, three or more linked children	148,775

Notes: Counts of person records, unweighted. Indented rows are subsets of the row they refer to; they are not successive exclusions, and the same-sex estimation sample is not obtained by subtracting the rows above it. The four sex compositions partition the same-sex estimation sample.

A.2 Housing outcomes

We measure three features of the dwelling, all reported once per household and assigned to every mother in it. The universe of each is housing units outside group quarters.

Rooms counts whole rooms used for living purposes, including kitchens and excluding bathrooms, porches, balconies, foyers, halls, and half-rooms. IPUMS codes valid counts from 1 to 27 and a top category of 30. In the 2009 file, 107 person records carry the value 28, which the codebook does not define; they are treated as missing. We cap the count at nine rooms because the Census Bureau imposed a national top code of nine rooms through the 2007 survey (IPUMS USA, 2026). Without the cap, a family’s measured size would depend on the survey year in which it was interviewed.

Bedrooms counts rooms that would be listed as bedrooms if the unit were for sale or rent. The IPUMS code minus one is the number of bedrooms; valid codes run from 1 (no bedroom) to 22. We cap the count at five, again because a national top code of five bedrooms applied through 2007.

Bedrooms are a component of rooms, but because the two counts are capped separately, capped rooms minus capped bedrooms is not a count of the other rooms in the dwelling.

Ownership equals one if the unit is owned or being bought, including units held with a mortgage, and zero if it is rented, including units occupied without payment of cash rent. Other codes are missing.

Every mother in both estimation samples has a valid tenure code and a valid bedroom count. One mother in the same-sex sample lacks a valid room count, which is why the room samples in that design are one observation smaller.

A.3 The two instruments

Same-age oldest pair. This design studies the move from one to two resident children among mothers whose oldest child is young. The sample consists of base-sample women with at least one linked child whose oldest child is aged 0 to 5. The instrument and the treatment are

$$Z_i^A = \mathbf{1}\{m_i \geq 2, a_{i1} = a_{i2}\}, \quad D_i^A = \mathbf{1}\{m_i \geq 2\}, \quad (1)$$

so the instrument indicates that the two oldest resident children are recorded at the same age and the treatment indicates a second resident child. Every mother with $Z_i^A = 1$ has $D_i^A = 1$ by construction. The event age is the oldest child's age, $e_i = a_{i1}$. If that child is the mother's first birth, e_i measures the years since that birth; the data do not confirm that it is.

The instrument is an age-only proxy for a twin birth, not a twin indicator, and we have not measured what share of same-age pairs are twins. Ages are recorded in whole years and our extract carries no quarter of birth, so two children born within a year of each other can share an age at the interview, and a step-sibling or adopted child of the same age produces the same record. Angrist and Evans (1998, p. 454) measure the resulting error in the 1980 Census: defining twins by age in years alone yields a second-birth twin rate 35 percent higher than defining them by age and quarter of birth, and for that reason they restrict their twins analysis to 1980. In our sample 1.9 percent of eligible mothers have a same-age pair, and 388 have three oldest children of the same age. Twin births are themselves not random: Angrist et al. (2010, p. 775) note that twinning varies with maternal age and race and that twin births change child spacing and child health.

Same-sex oldest pair. This design studies the move from two to three resident children. The sample consists of base-sample women with at least two linked children whose two oldest children have different ages, whose second-oldest child is aged 0 to 5, and whose two oldest children both have a recorded sex. The instrument and treatment are

$$Z_i^B = \mathbf{1}\{s_{i1} = s_{i2}\}, \quad D_i^B = \mathbf{1}\{m_i \geq 3\}, \quad (2)$$

where s_{ik} is the sex of the k th oldest linked child. The event age is the second child's age, $e_i = a_{i2}$, which proxies for the years since the second birth, when the instrument is realized. The requirement that the two oldest children differ in age removes the mothers who drive the first design and makes the order of the first two children well defined. The sample is not restricted to mothers with exactly two children and does not condition on the treatment. When the second and third children share an age, the child labeled second is arbitrary; Section A.6 drops these mothers.

The instrument predicts a third resident child, and its first stage is positive: mothers of two boys or two girls are more likely to have another child. Because the instrument is realized at the second birth, no event age in this design precedes the instrument, and event age 0 cannot serve as a period before exposure.

A.4 Estimating equations and inference

For each design, outcome Y_i , and window of event ages, we estimate a reduced form and a first stage on the same mothers:

$$Y_i = \delta Z_i + W_i' \gamma + u_i, \quad D_i = \pi Z_i + W_i' \kappa + v_i. \quad (3)$$

The reduced-form coefficient δ is the difference in housing between mothers with and without the sibling feature, and π is the corresponding difference in the probability of the additional child. The controls W_i are complete sets of indicators for the mother's age at the event (her age at the interview minus e_i), her race, the survey year, and the event age. We do not control for the mother's current marital status, education, or state of residence, all of which can respond to fertility; Angrist and Evans (1998, p. 465, footnote 12) exclude education and husband's earnings for the same reason. Regressions are weighted by the mother's ACS person weight. Standard errors are clustered by household, and each outcome is estimated on the mothers for whom it is observed.

With one instrument and one endogenous variable, two-stage least squares is the ratio of the two coefficients:

$$\hat{\beta} = \hat{\delta} / \hat{\pi}. \quad (4)$$

The ratio is a causal effect under three assumptions: the instrument is as good as randomly assigned given W_i ; it affects housing only through the additional child; and it moves every mother's fertility in the same direction. Under them, β is the average effect of the additional child for mothers whose fertility responds to the instrument. For the first design these are mothers who, without a same-age pair, would have one resident child at that event age. For the second they are mothers who have a third child because their first two share a sex. The two groups differ in age, in family size, and in the margin they cross, so the two estimates answer different questions. We do not test these assumptions, and no sign restriction on the housing effect is imposed or tested.

The baseline does not include separate indicators for the sex of the first and second child. Angrist and Evans (1998, pp. 460–461) add them because the same-sex indicator is an interaction of the two children's sexes: with slightly more boys than girls born, it is mildly correlated with each child's sex, and a child's sex may affect parents for reasons unrelated to family size. They report that their estimates are virtually unchanged by these controls (pp. 466–467). Section A.6 reports the same check.

Confidence intervals for the reduced form, first stage, and two-stage estimate are the estimate plus or minus 1.96 clustered standard errors. The first-stage F statistic is the household-clustered Wald statistic $(\hat{\pi} / \widehat{\text{se}}(\hat{\pi}))^2$. With a single instrument and a single endogenous variable, this is the statistic to which the Kleibergen–Paap robust F reduces (Andrews et al., 2018, pp. 16–17).

A large first stage protects against weak-instrument bias. It says nothing about the exclusion restriction. Because inference on $\hat{\beta}$ can still be unreliable when the first stage is imprecise, we also report the Anderson–Rubin test (Anderson and Rubin, 1949), which Andrews et al. (2018, p. 3) recommend whenever there is a single instrument. The hypothesis $\beta = \beta_0$ says that, net of the controls, the instrument has no effect on $Y_i - \beta_0 D_i$. Its coefficient on that variable is $\hat{\delta} - \beta_0 \hat{\pi}$, and the test statistic is

$$AR(\beta_0) = \frac{(\hat{\delta} - \beta_0 \hat{\pi})^2}{\hat{V}_\delta - 2\beta_0 \hat{C}_{\delta\pi} + \beta_0^2 \hat{V}_\pi}, \quad (5)$$

where $\hat{V}_\delta$, $\hat{V}_\pi$, and $\hat{C}_{\delta\pi}$ are the clustered variances and covariance of the reduced-form and first-stage coefficients. The statistic is compared with the 5 percent critical value of an F distribution with one numerator degree of freedom and the denominator degrees of freedom of the clustered regressions.

Under the maintained assumptions and the large-sample household-cluster approximation, this inference remains valid when the instrument is weak.

We evaluate the test on a finite grid of candidate values: from -3 to 3 in steps of 0.1 for rooms and bedrooms, and from -50 to 50 percentage points in steps of 2 for ownership. The tables report the smallest and largest grid values that are not rejected. In every cell with at least one accepted value, the accepted grid values form one unbroken run. Values between grid points were not evaluated, so the reported ranges describe the grid, not the exact endpoints of a continuous confidence set. A single entry in braces means that one tested value was accepted and its neighbors on the grid were rejected; it is not a zero-width interval. “None” means that no tested value was accepted.

A.5 Results

Table 2 reports the estimates pooled over event ages 0 to 5, and Table 3 reports them at event ages 3 and 5. Figure 1 plots all eighteen reduced forms.

Table 2: Pooled estimates, event ages 0–5

Outcome	Mothers	Households	Instrument equal to one	Reduced form	First stage (pp)	First-stage F	2SLS	AR accepted grid values
<i>A. Same-age oldest pair, event age 0–5</i>								
Rooms	857,607	855,366	16,496	0.329 (0.017)	66.21 (0.24)	77,396	0.496 (0.026)	{0.5}
Bedrooms	857,607	855,366	16,496	0.221 (0.010)	66.21 (0.24)	77,396	0.333 (0.014)	none
Owner (pp)	857,607	855,366	16,496	3.11 (0.46)	66.21 (0.24)	77,396	4.69 (0.70)	[4, 6]
<i>B. Same-sex oldest pair, event age 0–5</i>								
Rooms	656,985	656,159	329,974	−0.038 (0.005)	3.27 (0.12)	715	−1.149 (0.164)	[−1.4, −0.9]
Bedrooms	656,986	656,160	329,974	−0.034 (0.003)	3.27 (0.12)	715	−1.042 (0.096)	[−1.2, −0.9]
Owner (pp)	656,986	656,160	329,974	−0.25 (0.15)	3.27 (0.12)	715	−7.50 (4.51)	[−16, 0]

Notes: Each row reports the reduced form and first stage of equation (3) and the two-stage least-squares estimate of equation (4), with household-clustered standard errors in parentheses; 95 percent confidence intervals are the estimate ± 1.96 standard errors. The treatment is a second resident child in panel A and a third in panel B. Rooms and bedrooms are counts, capped at nine and five; ownership and the first stage are in percentage points. “AR accepted grid values” gives the smallest and largest grid values of the effect not rejected by the Anderson–Rubin test at the 5 percent level (grid in the text); braces denote a single accepted grid value and “none” that no grid value was accepted. Regressions weighted by the ACS person weight.

Same-age oldest pair. The first stage is large, 66 percentage points, because a same-age pair guarantees a second child: the coefficient is approximately one minus the probability that a comparable mother without such a pair already has a second resident child. Mothers with a same-age pair live in 0.33 more rooms (standard error 0.017) and 0.22 more bedrooms, and they are 3.1 percentage points more likely to own. Scaled by the first stage, a second child is associated with half a room, a third of a bedroom, and a 4.7-point higher ownership rate. The first stage falls from 52 points at event age 3 to 35 points at event age 5, as more mothers without a same-age pair have a second child, while the room and bedroom reduced forms change little. The two-stage estimates

therefore rise with the event age, to 1.1 rooms and 15 points of ownership at age 5. At event age 3 the ownership reduced form is 1.6 points and its confidence interval includes zero. These magnitudes carry the qualifications above: the proxy mixes twins with closely spaced and blended sibling pairs in an unmeasured proportion, and twin births are associated with maternal characteristics that may affect housing directly.

Same-sex oldest pair. The first stage is 3.3 percentage points pooled over event ages 0 to 5 and rises to 4.7 points at event age 3 and 6.0 points at event age 5. With more than 650,000 mothers it is precisely estimated, with an F statistic of 715. The pooled figure is smaller than the 6.2 points that Angrist and Evans (1998, p. 461) estimate in the 1980 Census, but the samples differ: ours ends five years after the second child’s birth, whereas theirs includes mothers whose second child is up to seventeen, and the two censuses are a generation apart. Mothers whose two oldest children share a sex live in 0.038 fewer rooms (standard error 0.005) and 0.034 fewer bedrooms (0.003), and they are 0.25 percentage points less likely to own, a difference whose confidence interval includes zero. The room and bedroom differences are close in magnitude, but because the two counts are capped separately we do not read their difference as a statement about other rooms, and we do not test them against each other.

The two-stage estimate for rooms is -1.15 , and the Anderson–Rubin test accepts grid values from -1.4 to -0.9 . It is large because the reduced form, though small, is divided by a first stage of 0.033: $-0.0376/0.0327 = -1.15$. Read under the assumptions of Section A.4, a third child reduces the family’s dwelling by more than a full room. The data alone cannot tell this reading apart from a failure of the exclusion restriction. A genuine negative effect is possible if a larger family has less to spend on housing, or delays a move it would otherwise make. Alternatively, sibling sex may affect housing directly. Under the exclusion restriction the reduced form equals $\pi\beta$. Any direct effect of sibling sex on housing adds to the reduced form in full while leaving the first stage unchanged, so it enters the two-stage estimate multiplied by $1/\pi$, about 31. Jones (2015) shows how direct effects on mothers whose fertility does not respond to the instrument bias the instrumental-variable estimand. Bedrooms are a natural candidate channel, since two children of the same sex can share a bedroom more easily than a boy and a girl. Angrist et al. (2010, p. 775) raise economies of scale through room or clothes sharing as a potential violation of the same-sex exclusion restriction, attributing the concern to Rosenzweig and Wolpin (2000). In Israeli subsamples in which sex composition does not predict family size, they find no reduced-form relation between sex composition and their outcomes, which concern the children’s schooling and later lives rather than housing (p. 799). A negative bedroom reduced form is consistent with the room-sharing hypothesis, but it does not identify it.

A.6 Checks on the same-sex design

We ran four checks on the same-sex estimates, all on the estimation sample and with the specification of equation (3) unless stated. They ask whether the negative housing difference depends on how the instrument is coded, which sibling compositions carry it, and when it appears after the second child’s birth. They check coding, measurement, and sensitivity. None of them tests the exclusion restriction, and none certifies a causal reading in either direction. Table 4 and Figure 2 report the pooled estimates; Table 5 and Figure 3 report them by the age of the second child; Table 6 reports unadjusted means.

Sex of each child. Adding separate indicators for a boy first and a boy second, as Angrist and Evans (1998) do, leaves the estimates essentially unchanged. The room reduced form moves from

Table 3: Estimates at event ages 3 and 5

Outcome	Mothers	Households	Instrument equal to one	Reduced form	First stage (pp)	First-stage F	2SLS	AR accepted grid values
<i>A. Same-age oldest pair, event age 3</i>								
Rooms	147,986	147,922	2,726	0.332 (0.042)	52.46 (0.24)	49,032	0.633 (0.080)	[0.5, 0.7]
Bedrooms	147,986	147,922	2,726	0.212 (0.023)	52.46 (0.24)	49,032	0.405 (0.043)	{0.4}
Owner (pp)	147,986	147,922	2,726	1.63 (1.14)	52.46 (0.24)	49,032	3.11 (2.17)	[0, 6]
<i>A. Same-age oldest pair, event age 5</i>								
Rooms	142,101	142,039	2,498	0.382 (0.041)	34.83 (0.22)	24,972	1.096 (0.119)	[0.9, 1.3]
Bedrooms	142,101	142,039	2,498	0.234 (0.023)	34.83 (0.22)	24,972	0.671 (0.067)	[0.6, 0.8]
Owner (pp)	142,101	142,039	2,498	5.26 (1.15)	34.83 (0.22)	24,972	15.11 (3.30)	[10, 20]
<i>B. Same-sex oldest pair, event age 3</i>								
Rooms	108,847	108,825	54,561	-0.051 (0.012)	4.67 (0.35)	174	-1.089 (0.284)	[-1.6, -0.6]
Bedrooms	108,847	108,825	54,561	-0.040 (0.007)	4.67 (0.35)	174	-0.856 (0.165)	[-1.2, -0.6]
Owner (pp)	108,847	108,825	54,561	-0.53 (0.36)	4.67 (0.35)	174	-11.31 (7.82)	[-26, 2]
<i>B. Same-sex oldest pair, event age 5</i>								
Rooms	92,246	92,229	46,201	-0.053 (0.013)	6.04 (0.41)	214	-0.874 (0.235)	[-1.3, -0.5]
Bedrooms	92,247	92,230	46,201	-0.045 (0.007)	6.04 (0.41)	214	-0.745 (0.137)	[-1.0, -0.5]
Owner (pp)	92,247	92,230	46,201	-0.33 (0.40)	6.04 (0.41)	214	-5.41 (6.59)	[-18, 6]

Notes: See Table 2. Event age is the oldest child's age in panel A and the second-oldest child's age in panel B, in whole years at the interview.

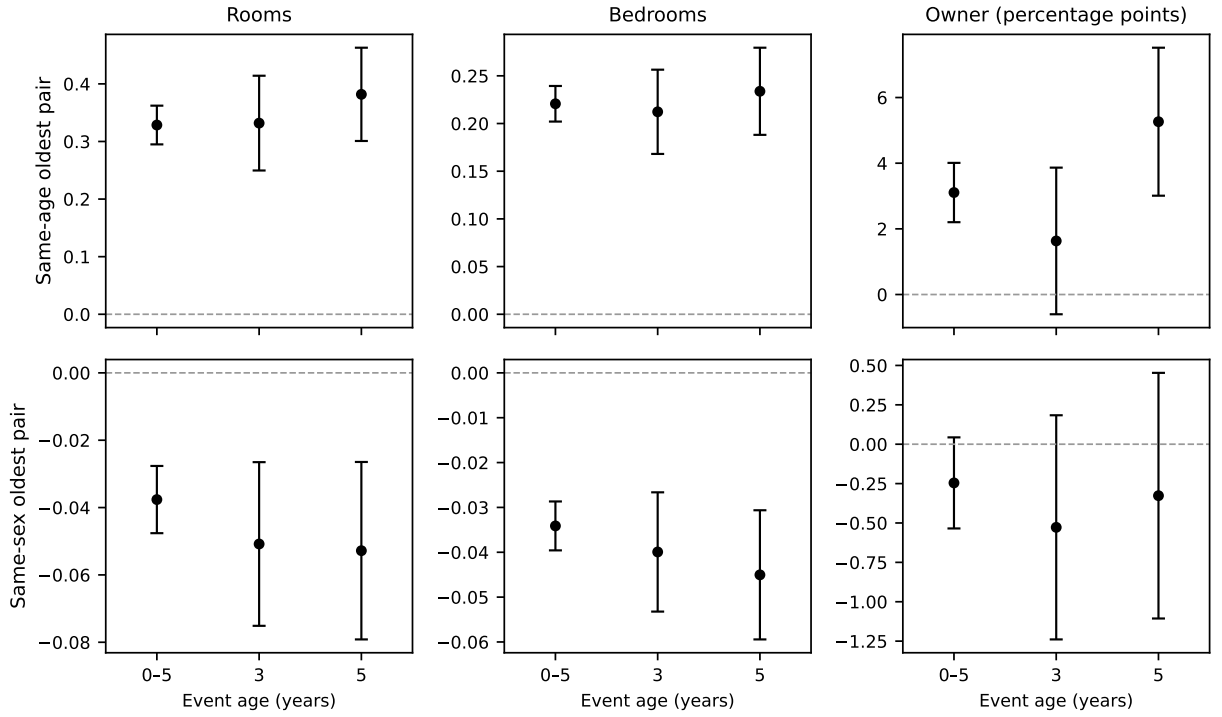


Figure 1: Reduced-form differences in housing by sibling composition

Notes: Each point is the reduced-form coefficient δ of equation (3), with a 95 percent confidence interval from household-clustered standard errors, for event ages 0–5 pooled and for event ages 3 and 5. Top row: mothers whose two oldest linked children are the same age, relative to mothers whose oldest child is the same age but who have no same-age pair. Bottom row: mothers whose two oldest linked children have the same sex, relative to mothers of one boy and one girl. Ownership in percentage points.

−0.0376 to −0.0380, the bedroom reduced form from −0.0341 to −0.0342, and the first stage stays at 3.27 percentage points.

Tied second and third children. In 11,500 estimation-sample mothers the second and third linked children are recorded at the same age, making the sex of the child labeled second ambiguous while leaving the event age unchanged. Dropping these mothers also leaves the estimates essentially unchanged: −0.0378 for rooms, −0.0347 for bedrooms, and a first stage of 3.27 points on 645,486 mothers.

Two boys and two girls. Replacing the same-sex indicator with separate indicators for two boys and two girls, both measured against the 327,012 mothers of one boy and one girl, shows that both compositions carry the negative difference. For rooms the coefficients are −0.0306 (standard error 0.0061) for two boys and −0.0454 (0.0063) for two girls; for bedrooms they are −0.0322 (0.0033) and −0.0363 (0.0035). The first stages are very similar, 3.28 and 3.27 points. The difference between the two boys and two girls coefficients, with a standard error from their joint covariance, is 0.0148 rooms (0.0072), with a 95 percent confidence interval of [0.0006, 0.0289]. For bedrooms it is 0.0041 (0.0039) and for ownership 0.09 points (0.21), both with intervals that include zero. The room difference is one of several contrasts we examined, is not adjusted for multiple comparisons, and lies close to the conventional threshold; we do not draw conclusions from it.

Age of the second child. At event age 0 the first stage is essentially zero, 0.09 percentage points ($F = 0.9$), and the room and bedroom reduced forms are small, with confidence intervals that include zero. This is the first year of exposure to the instrument, not a period before it. The estimated third-child difference is small and imprecise, and little time has passed for housing to adjust, so the small differences are not evidence that sibling sex has no direct effect. By event age 1 the bedroom reduced form is already −0.0278 (0.0066), while the first stage is 1.20 points. From age 1 to age 5 the first stage rises about fivefold, to 6.04 points, whereas the bedroom reduced form rises about 1.6-fold, to −0.0450. With a constant effect of the third child and no direct effect, the two would move proportionally across ages. We read the gap only descriptively. Each event age is a different cross-section of mothers, not the same families followed over time, and both the composition of mothers and the effect of a third child may differ across ages. The ownership reduced form is imprecise at every age, and every one of its confidence intervals includes zero.

Unadjusted means. Table 6 reports the number of mothers and the weighted and unweighted means of the treatment and the three outcomes for each of the four sex compositions. The ordering matches the regressions. Mothers of two boys or two girls are more likely to have a third resident child (0.245 and 0.246, weighted) than mothers of a boy and a girl (0.212 and 0.214). They also live in slightly fewer rooms and bedrooms. Ownership rates differ little across the four groups.

Summary. The two designs measure different objects, and their estimates should not be averaged or set against each other. Under its assumptions, the same-age design says that a second young child is associated with more space and a higher likelihood of owning. The same-sex design yields a negative estimate for a third child. The negative association persists after adding separate controls for the sex of each child, excluding tied second and third ages, and separating two-boy and two-girl families; the limitations of the birth-order and age-only proxies are unchanged. Whether it reflects an effect of the third child or a direct effect of sibling composition on housing is not identified by these data.

Table 4: Same-sex design: alternative specifications, event ages 0–5

Specification	Reduced form			First stage	
	Rooms	Bedrooms	Owner (pp)	(pp)	Mothers
Baseline	−0.0376 (0.0051)	−0.0341 (0.0028)	−0.25 (0.15)	3.27 (0.12)	656,986
With first- and second-child sex	−0.0380 (0.0051)	−0.0342 (0.0028)	−0.25 (0.15)	3.27 (0.12)	656,986
Excluding tied second and third ages	−0.0378 (0.0051)	−0.0347 (0.0028)	−0.23 (0.15)	3.27 (0.12)	645,486
Two boys vs. mixed	−0.0306 (0.0061)	−0.0322 (0.0033)	−0.20 (0.18)	3.28 (0.15)	656,986
Two girls vs. mixed	−0.0454 (0.0063)	−0.0363 (0.0035)	−0.29 (0.18)	3.27 (0.15)	656,986
Two boys minus two girls	0.0148 (0.0072)	0.0041 (0.0039)	0.09 (0.21)	0.01 (0.18)	656,986

Notes: Reduced-form and first-stage coefficients from equation (3), household-clustered standard errors in parentheses. “With first- and second-child sex” adds indicators for a boy first and a boy second. “Excluding tied second and third ages” drops mothers whose second and third linked children are recorded at the same age. The two-boys and two-girls rows are the coefficients on separate indicators in a single regression, relative to mothers of one boy and one girl; their first stage is the corresponding coefficient in the first-stage regression. The last row is the difference between them, with a standard error from their joint clustered covariance. The mothers column refers to bedrooms and ownership; room samples are one mother smaller. Ownership in percentage points.

Table 5: Same-sex design by age of the second child

Age of second child (years)	First stage (pp)	First-stage F	Reduced form			Mothers
			Rooms	Bedrooms	Owner (pp)	
0	0.09 (0.10)	0.9	−0.0086 (0.0120)	−0.0037 (0.0066)	−0.20 (0.34)	120,739
1	1.20 (0.20)	35.7	−0.0149 (0.0121)	−0.0278 (0.0066)	0.04 (0.35)	119,153
2	3.10 (0.30)	108.8	−0.0455 (0.0122)	−0.0403 (0.0066)	−0.59 (0.35)	115,894
3	4.67 (0.35)	173.6	−0.0508 (0.0124)	−0.0399 (0.0068)	−0.53 (0.36)	108,847
4	5.58 (0.39)	206.1	−0.0607 (0.0129)	−0.0535 (0.0071)	0.19 (0.38)	100,106
5	6.04 (0.41)	213.6	−0.0528 (0.0134)	−0.0450 (0.0073)	−0.33 (0.40)	92,247

Notes: Reduced form and first stage of equation (3) estimated separately at each age of the second child, household-clustered standard errors in parentheses. The first stage and mothers columns refer to bedrooms and ownership; at age 5 the room sample is one mother smaller. Ownership in percentage points.

Table 6: Same-sex design: means by sex of the two oldest children

Two oldest children	Mothers	Weighted (millions)	Three or more children	Rooms	Bedrooms	Owner
<i>A. Weighted means</i>						
Two boys	173,813	19.41	0.245	5.835	2.961	0.537
Two girls	156,161	17.64	0.246	5.810	2.953	0.532
Boy, then girl	164,253	18.51	0.212	5.857	2.988	0.535
Girl, then boy	162,759	18.36	0.214	5.860	2.993	0.538
<i>B. Unweighted means</i>						
Two boys	173,813	19.41	0.242	6.000	3.021	0.599
Two girls	156,161	17.64	0.243	5.986	3.021	0.596
Boy, then girl	164,253	18.51	0.209	6.023	3.051	0.596
Girl, then boy	162,759	18.36	0.211	6.029	3.055	0.599

Notes: Same-sex estimation sample, event ages 0–5. Weighted means use the ACS person weight, and the weighted count is the sum of weights across the annual samples, not a single-year population estimate. “Three or more children” is the share with three or more linked resident children. Rooms and bedrooms are capped at nine and five; ownership is a share. Means are not adjusted for controls.

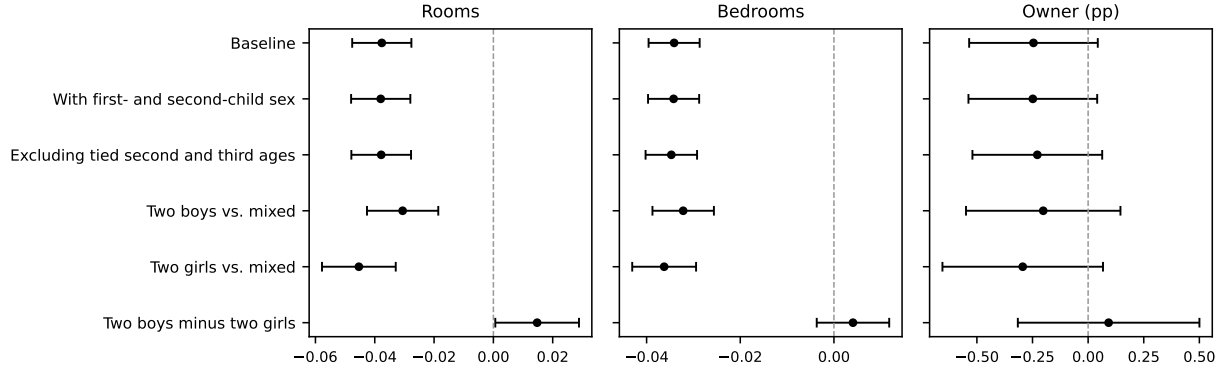


Figure 2: Same-sex design: reduced forms across specifications

Notes: Reduced-form coefficients from Table 4 with 95 percent confidence intervals from household-clustered standard errors. Ownership in percentage points.

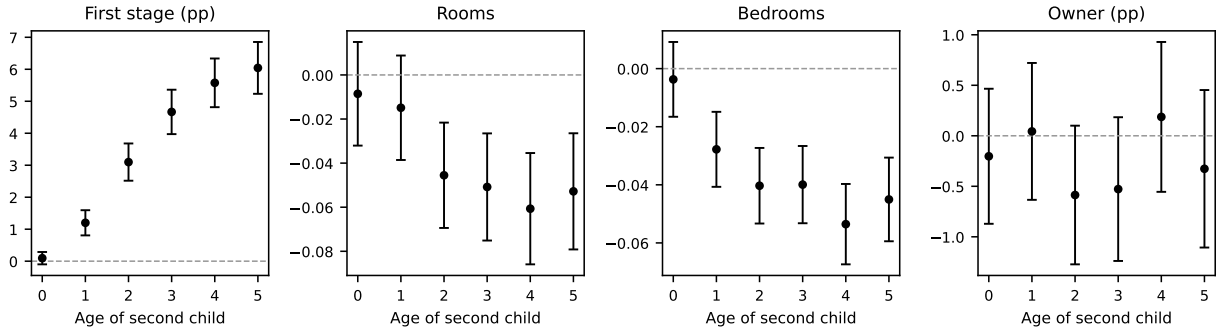


Figure 3: Same-sex design by age of the second child

Notes: First stage (percentage points of a third resident child) and reduced forms from Table 5, with 95 percent confidence intervals from household-clustered standard errors. Each age is a separate cross-section of mothers.

References

- T. W. Anderson and Herman Rubin. Estimation of the parameters of a single equation in a complete system of stochastic equations. *Annals of Mathematical Statistics*, 20(1):46–63, 1949.
- Isaiah Andrews, James H. Stock, and Liyang Sun. Weak instruments in IV regression: Theory and practice. Manuscript, November 20, 2018. Published in *Annual Review of Economics* 11 (2019): 727–753, 2018.
- Joshua Angrist, Victor Lavy, and Analia Schlosser. Multiple experiments for the causal link between the quantity and quality of children. *Journal of Labor Economics*, 28(4):773–823, 2010.
- Joshua D. Angrist and William N. Evans. Children and their parents’ labor supply: Evidence from exogenous variation in family size. *American Economic Review*, 88(3):450–477, 1998.
- IPUMS USA. Variable descriptions and codes: MOMLOC, RELATE, ROOMS, BEDROOMS, OWNERSHP, PERWT. <https://usa.ipums.org/usa-action/variables/>, 2026. University of Minnesota. Accessed September 22, 2026.
- Damon Jones. The economics of exclusion restrictions in IV models. Working Paper 21391, National Bureau of Economic Research, 2015.
- Mark R. Rosenzweig and Kenneth I. Wolpin. Natural “natural experiments” in economics. *Journal of Economic Literature*, 38(4):827–874, 2000.